Chapter 37 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

Why do many trees begin development with a taproot and after one or two years change to a fibrous root system with many horizontal surface roots and only a few vertical, deep anchoring roots? **(BIG IDEA 2)**

Concept 37.1: Soil contains a living, complex ecosystem

- Soil particles of various sizes derived from the breakdown of rock are found in soil. Soil particle size affects the availability of water, oxygen, and minerals in the soil.
- A soil's composition encompasses its inorganic and organic components. **Topsoil** is a complex ecosystem teeming with bacteria, fungi, protists, animals, and the roots of plants.
- Some agricultural practices can deplete the mineral content of soil, tax water reserves, and promote erosion. The goal of soil conservation is to minimize this damage.

How is soil a complex ecosystem?

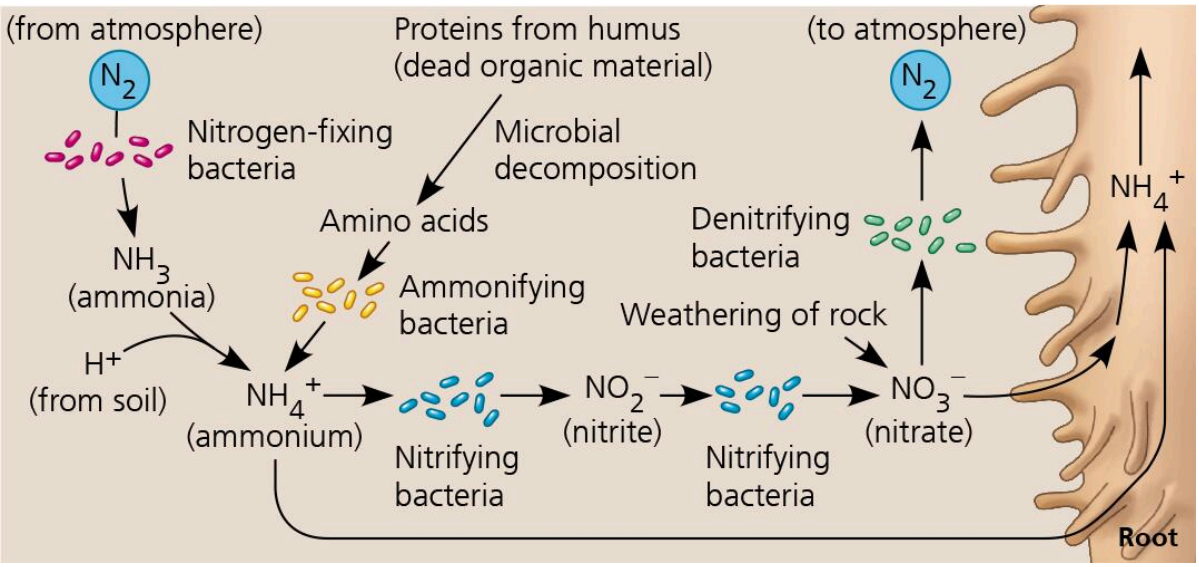
Concept 37.2: Plant roots absorb many types of essential elements from the soil

- **Macronutrients**, elements required in relatively large amounts, include carbon, oxygen, hydrogen, nitrogen, and other major ingredients of organic compounds. **Micronutrients**, elements required in very small amounts, typically have catalytic functions as cofactors of enzymes.
- Deficiency of a mobile nutrient usually affects older organs more than younger ones; the reverse is true for nutrients that are less mobile within a plant. Macronutrient deficiencies are most common, particularly deficiencies of nitrogen, phosphorus, and potassium.
- Rather than tailoring the soil to match the plant, genetic engineers are tailoring the plant to match the soil.

Do plants need soil to grow? Explain.

Concept 37.3: Plant nutrition often involves relationships with other organisms

- **Rhizobacteria** derive their energy from the **rhizosphere**, a microorganism-enriched ecosystem intimately associated with roots. Plant secretions support the energy needs of the rhizosphere. Some rhizobacteria produce antibiotics, whereas others make nutrients more available for plants. Most are free-living, but some live inside plants. Plants satisfy most of their huge need for nitrogen from the bacterial decomposition of **humus** and the fixation of gaseous nitrogen.



In soil, N 2 is processed by nitrogen-fixing bacteria that produce N H 3 (ammonia). H plus (from soil) is added to make N H 4 superscript plus (ammonium). Proteins from humus (dead organic material) go through microbial decomposition, break into amino acids, the ammonifying bacteria produce N H 4 superscript plus, as well. N H 4 superscript plus is taken up by roots, or nitrifying bacteria produce N O 2 superscript minus (nitrite) that is processed by nitrifying bacteria again to become N O 3 superscript minus (nitrate). Weathering of rock also produces this. Denitrifying bacteria then produce N 2 that is released from the soil. From the root, N H 4 superscript plus, nitrate, and nitrogenous organic composes are exported in the xylem to the shoot system.

- Nitrogen-fixing bacteria convert atmospheric N2 to nitrogenous minerals that plants can absorb as a nitrogen source for organic synthesis. The most efficient mutualism between plants and nitrogen-fixing bacteria occurs in the **nodules** that are formed by *Rhizobium* bacteria growing in the roots of legumes. These bacteria obtain sugar from the plant and supply the plant with fixed nitrogen. In agriculture, legume crops are rotated with other crops to restore nitrogen to the soil.
- **Mycorrhizae** are mutualistic associations of fungi and roots. The fungal hyphae of mycorrhizae absorb water and minerals, which they supply to their plant hosts.
- **Epiphytes** grow on the surfaces of other plants but acquire water and minerals from rain. Parasitic plants absorb nutrients from host plants. Carnivorous plants supplement their mineral nutrition by digesting animals.

Do all plants gain energy directly from photosynthesis? Explain.